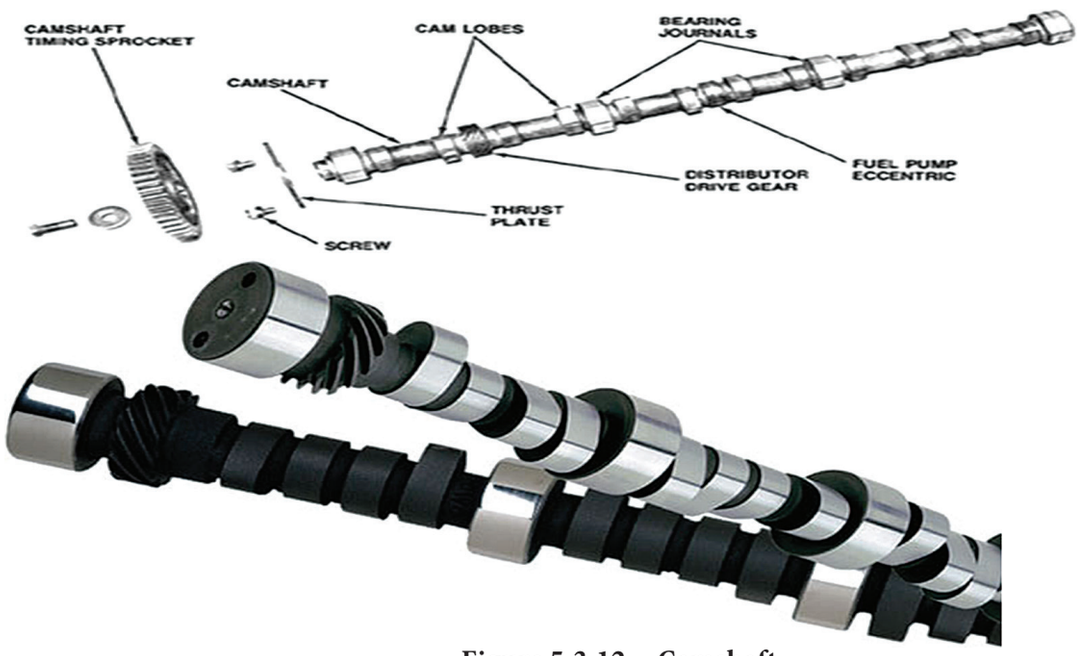


Figure 5.3.12 Camshaft

cam shaft performs the function of actuating the petrol pump and also the skew gear arrangement which is required for operating the oil pump. The timing gear is fitted at the edge of the cam shaft. This gear arrangement is placed in the cylinder block. This gear is made up of special steel for avoiding wear and tear. Cam shaft is actuated by this timing gear with the help of crank shaft. Cam shaft rotates always at half the speed of crank shaft rotation.

5.3.13 Valve

In the engines (generally in four stroke engines) valves are used for supplying enough air fuel mixture into the combustion chamber for combustion and for expelling the burned gases to the exhaust by opening and closing of ports in the engine. There are two valves namely intake and exhaust valves present in the engines. These valves are made up of nickel, chromium alloy steel or silicon chromium alloy steel. The head of the intake valve is generally larger in size than the exhaust valve. These valves are made up of the process called drop forging. In the present days, exhaust valves are produced by austenitic steel. Generally poppet valves are used in present engines.

5.3.13.1 Methods of operating the valves:

In the engine, operating mechanism of the valve varies with the position of the valve arrangements. Valves are placed on the cylinder head in such a way that the valve is moved downwards to open the ports. In another system, valve is placed at the cylinder head in such a way to move upwards to open the ports. Valve mechanisms are generally classified into,

a. Over head poppet valve mechanism
b. Straight poppet valve mechanism

a) Over head poppet valve mechanism:

Construction

This consists of two moving parts, namely push rod and rocker arm. Cam in the can shaft is always in touch with the tappet. One end of the push rod is in contact with tappet and other end with the rocker arm. The other end of

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